

Problem - Set - 8 - Solution

2) Given

$$H_0 = \begin{bmatrix} E_1 & 0 \\ 0 & E_2 \end{bmatrix}, V(t) = \begin{bmatrix} 0 & \frac{\hbar \Omega}{2} e^{i\omega t} \\ \frac{\hbar \Omega^*}{2} e^{-i\omega t} & 0 \end{bmatrix}$$

a) Solution

$$U_0(t) = e^{-\frac{i}{\hbar} H_0 t}$$

$$U_0(t) = \begin{bmatrix} e^{-\frac{i E_1 t}{\hbar}} & 0 \\ 0 & e^{-\frac{i E_2 t}{\hbar}} \end{bmatrix}$$

$$U_0^\dagger(t) = \begin{bmatrix} e^{i \frac{E_1 t}{\hbar}} & 0 \\ 0 & e^{i \frac{E_2 t}{\hbar}} \end{bmatrix}$$

$$U(t) = U_0^\dagger(t) V(t) U_0(t)$$

$$= \begin{bmatrix} e^{i \frac{E_1 t}{\hbar}} & 0 \\ 0 & e^{i \frac{E_2 t}{\hbar}} \end{bmatrix} \begin{bmatrix} 0 & \frac{\hbar \Omega}{2} e^{i\omega t} \\ \frac{\hbar \Omega^*}{2} e^{-i\omega t} & 0 \end{bmatrix} \begin{bmatrix} e^{-\frac{i E_1 t}{\hbar}} & 0 \\ 0 & e^{-\frac{i E_2 t}{\hbar}} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & e^{i \frac{E_1 t}{\hbar}} \frac{\hbar \Omega}{2} e^{i\omega t} e^{-\frac{i E_2 t}{\hbar}} \\ e^{i \frac{E_2 t}{\hbar}} \frac{\hbar \Omega^*}{2} e^{-i\omega t} e^{-\frac{i E_1 t}{\hbar}} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \frac{\hbar \Omega}{2} e^{i\omega t} e^{i \frac{(E_1 - E_2)t}{\hbar}} \\ \frac{\hbar \Omega^*}{2} e^{-i\omega t} e^{i \frac{(E_2 - E_1)t}{\hbar}} & 0 \end{bmatrix}$$

Where, $\omega_0 = \frac{E_2 - E_1}{\hbar}$

$$\Delta = \omega - \omega_0$$

2)

$$e^{i\omega t} e^{i \underbrace{(E_1 - E_2)}_{= -\omega_0} t} = e^{i\omega t} e^{-i\omega_0 t} = e^{i(\omega - \omega_0)t} = e^{i\Delta t}$$

$$e^{-i\omega t} e^{i \underbrace{(E_2 - E_1)}_{= \omega_0} t} = e^{-i\omega t} e^{i\omega_0 t} = e^{-i(\omega - \omega_0)t} = e^{-i\Delta t}$$

$$V_D(t) = \frac{\hbar}{2} \begin{bmatrix} 0 & \Omega e^{i\Delta t} \\ \Omega^* e^{-i\Delta t} & 0 \end{bmatrix}$$

b) Solution

For two-level system we can use 12) and 22)

SE in IP

$$|\psi_I(t)\rangle = c_1(t)|1\rangle + c_2(t)|2\rangle$$

SE in time-dependent SE

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = V_D(t) |\psi_I(t)\rangle$$

$$i\hbar \frac{d}{dt} (c_1(t)|1\rangle + c_2(t)|2\rangle) = V_D(t) |\psi_I(t)\rangle$$

$$i\hbar \left[\frac{d}{dt} [c_1(t)] |1\rangle + \frac{d}{dt} [c_2(t)] |2\rangle \right] = V_D(t) |\psi_I(t)\rangle$$

$$i\hbar \left[\dot{c}_1(t) |1\rangle + \dot{c}_2(t) |2\rangle \right] = V_D(t) |\psi_I(t)\rangle$$

$$i\hbar \begin{bmatrix} \dot{c}_1(t) \\ \dot{c}_2(t) \end{bmatrix} = V_2(t) \begin{bmatrix} c_1(t) \\ c_2(t) \end{bmatrix}$$

$$i\hbar \begin{bmatrix} \dot{c}_1(t) \\ \dot{c}_2(t) \end{bmatrix} = \frac{\hbar}{2} \begin{bmatrix} 0 & \Omega e^{i\Delta t} \\ \Omega^* e^{-i\Delta t} & 0 \end{bmatrix} \begin{bmatrix} c_1(t) \\ c_2(t) \end{bmatrix}$$

$$i \begin{bmatrix} \dot{c}_1(t) \\ \dot{c}_2(t) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} c_2(t) \Omega e^{i\Delta t} \\ c_1(t) \Omega^* e^{-i\Delta t} \end{bmatrix}$$

$$i\dot{c}_1(t) = \frac{\Omega}{2} c_2(t) e^{i\Delta t} \Rightarrow \dot{c}_1(t) = -\frac{i}{2} \Omega e^{i\Delta t} c_2(t)$$

$$i\dot{c}_2(t) = \frac{\Omega^*}{2} c_1(t) e^{-i\Delta t} \Rightarrow \dot{c}_2(t) = -\frac{i}{2} \Omega^* e^{-i\Delta t} c_1(t)$$

Derive the differential equation for $c_2(t)$

$$\frac{d}{dt} [\dot{c}_2(t)] = \frac{d}{dt} \left[-\frac{i}{2} \Omega^* e^{-i\Delta t} c_1(t) \right]$$

$$\ddot{c}_2(t) = -\frac{i}{2} \Omega^* \frac{d}{dt} [e^{-i\Delta t} c_1(t)]$$

$$= -\frac{i}{2} \Omega^* \left[\frac{d}{dt} [e^{-i\Delta t}] c_1(t) + e^{-i\Delta t} \frac{d}{dt} [c_1(t)] \right]$$

$$= -\frac{i}{2} \Omega^* \left[(-i\Delta) e^{-i\Delta t} c_1(t) + \dot{c}_1(t) e^{-i\Delta t} \right]$$

$$= (i)^2 \frac{\Omega^*}{2} \Delta c_1(t) e^{-i\Delta t} + \frac{(i)^2}{4} \Omega^* \Omega e^{i\Delta t} c_2(t) e^{-i\Delta t}$$

$$= -\frac{\Omega^*}{2} \Delta c_1(t) e^{-i\Delta t} - \frac{1}{4} \Omega^* \Omega c_2(t) e^{i\Delta t} e^{-i\Delta t}$$

$$= -\frac{\Omega^*}{2} \Delta c_1(t) e^{-i\Delta t} - \frac{|\Omega|^2}{4} c_2(t) e^{(i\Delta t - i\Delta t)}$$

3)

$$= -\frac{\Omega^*}{2} \Delta C_1(t) e^{-i\Delta t} - \frac{|\Omega|^2}{4} C_2(t) e^0$$

$$= -\frac{\Omega^*}{2} \Delta C_1(t) e^{-i\Delta t} - \frac{|\Omega|^2}{4} C_2(t) \cdot 1$$

$$= -\frac{\Omega^*}{2} \Delta C_1(t) e^{-i\Delta t} - \frac{|\Omega|^2}{4} C_2(t)$$

$$= -\Delta \underbrace{\frac{\Omega^*}{2} C_1(t) e^{-i\Delta t}}_{= i \dot{C}_2(t)} - \frac{|\Omega|^2}{4} C_2(t)$$

$$= -\Delta (i \dot{C}_2(t)) - \frac{|\Omega|^2}{4} C_2(t)$$

$$= -i \Delta \dot{C}_2(t) - \frac{|\Omega|^2}{4} C_2(t)$$

$$\ddot{C}_2(t) = -i \Delta \dot{C}_2(t) - \frac{|\Omega|^2}{4} C_2(t)$$

c) Solution

$$\ddot{C}_2(t) = -i \Delta \dot{C}_2(t) - \frac{|\Omega|^2}{4} C_2(t), \quad \Delta = \omega - \frac{E_2 - E_1}{\hbar}$$

At initial time both systems are at 1st,

$$\left. \begin{aligned} C_2(0) &= 1 \\ C_1(0) &= 0 \end{aligned} \right\}$$

$$P_h \frac{d}{dt} \begin{bmatrix} c_1(t) \\ c_2(t) \end{bmatrix} = V_D(t) \begin{bmatrix} c_1(t) \\ c_2(t) \end{bmatrix}$$

At the initial condition, $t=0$

$$\left. \begin{aligned} c_1(0) &= 1 \\ c_2(0) &= 0 \end{aligned} \right\}$$

$$P_h \frac{d}{dt} \begin{bmatrix} c_1(0) \\ c_2(0) \end{bmatrix} = \frac{h}{2} \begin{bmatrix} 0 & \frac{h\omega}{2} e^{i\omega \cdot 0} \\ \frac{h\omega}{2} e^{-i\omega \cdot 0} & 0 \end{bmatrix} \begin{bmatrix} c_1(0) \\ c_2(0) \end{bmatrix}$$

$$= \frac{h}{2} \begin{bmatrix} 0 & \frac{h\omega}{2} \\ \frac{h\omega}{2} & 0 \end{bmatrix} \begin{bmatrix} c_1(0) \\ c_2(0) \end{bmatrix}$$

$$= \frac{h}{2} \begin{bmatrix} 0 & h\omega \\ h\omega & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{h}{2} \begin{bmatrix} 0 \\ h\omega \end{bmatrix}$$

$$\begin{bmatrix} i\dot{c}_1(0) \\ i\dot{c}_2(0) \end{bmatrix} = \frac{h}{2} \begin{bmatrix} 0 \\ h\omega \end{bmatrix}$$

$$\left. \begin{aligned} \dot{c}_1(0) &= 0 \\ \dot{c}_2(0) &= -i \frac{h\omega}{2} \end{aligned} \right\}$$



The General Solution

$$c_2(t) = e^{-\frac{i\omega t}{2}} b(t)$$

$$\frac{d}{dt} [c_2(t)] = \frac{d}{dt} \left[e^{-i\frac{\Delta t}{2}} b(t) \right]$$

$$= e^{-i\frac{\Delta t}{2}} \left(\dot{b} - i\frac{\Delta}{2} b \right)$$

$$\dot{c}_2(t) = e^{-i\frac{\Delta t}{2}} \left(\dot{b} - i\frac{\Delta}{2} b \right) \quad \dots \quad (1)$$

$$\frac{d}{dt} [\dot{c}_2(t)] = \frac{d}{dt} \left[e^{-i\frac{\Delta t}{2}} \left(\dot{b} - i\frac{\Delta}{2} b \right) \right]$$

$$\ddot{c}_2(t) = e^{-i\frac{\Delta t}{2}} \left(\ddot{b} - i\Delta \dot{b} - \frac{\Delta^2}{4} b \right) \quad \dots \quad (2)$$

Insert into the equation

$$\ddot{c}_2(t) = -i\Delta \dot{c}_2(t) - \frac{|\Omega|^2}{4} c_2(t)$$

and, cancel the common factor, $e^{-i\frac{\Delta t}{2}}$

$$\ddot{b} - i\Delta \dot{b} - \frac{\Delta^2}{4} b = -i\Delta \left(\dot{b} - i\frac{\Delta}{2} b \right) - \frac{|\Omega|^2}{4} b$$

$$= -i\Delta \dot{b} + i^2 \frac{\Delta^2}{2} b - \frac{|\Omega|^2}{4} b$$

$$= -i\Delta \dot{b} - \frac{\Delta^2}{2} b - \frac{|\Omega|^2}{4} b$$

$$\cancel{\ddot{b}} - \cancel{i\Delta \dot{b}} - \frac{\Delta^2}{4} b = -\cancel{i\Delta \dot{b}} - \frac{\Delta^2}{2} b - \frac{|\Omega|^2}{4} b$$

$$\ddot{b} + \frac{\Delta^2}{4} b + \frac{|\Omega|^2}{4} b = 0$$

$$\ddot{b} + \left(\frac{\Delta^2 + |\Omega|^2}{4} \right) b = 0$$



Where,

$$\Delta^2 + |S|^2 = \omega_R^2 \rightarrow \text{Rabi frequency}$$

$$\ddot{b} + \left(\frac{\omega_R^2}{4} \right) b = 0$$

from, SHO, solution,

$$b(t) = A \sin\left(\frac{\omega_R t}{2}\right) + B \cos\left(\frac{\omega_R t}{2}\right)$$

At initial condition,

$$C_1 = 1 = A$$

$$C_2 = 0 = B$$

$$b(t) = A \sin\left(\frac{\omega_R t}{2}\right)$$

$$\dot{C}_2(0) = -i \frac{\Omega^*}{2}$$

$$\dot{C}_2 = e^{-i \frac{\Delta t}{2}} \left(\dot{b} - i \frac{\Delta}{2} b \right)$$

At $t=0$, $b(0)=0$, so

$$\dot{C}_2(0) = \dot{b}(0) = A \frac{\omega_R}{2}$$

$$\dot{C}_2(0) = A \frac{\omega_R}{2} = -i \frac{\Omega^*}{2}$$

$$A = \frac{-i \Omega^*}{\omega_R}$$

$$C_2(t) = e^{-i \frac{\Delta t}{2}} b(t) = \frac{-i \Omega^*}{\omega_R} e^{-i \frac{\Delta t}{2}} \sin\left(\frac{\omega_R t}{2}\right)$$

7)

The transition Probability

$$P_{1 \rightarrow 2}(t) = |c_2(t)|^2$$

$$= \left| \frac{-i s_2^*}{\omega_R} \right|^2 \left| \sin\left(\frac{\omega_R t}{2}\right) \right|^2$$

$$= \frac{|s_2|^2}{\omega_R^2} \sin^2\left(\frac{\omega_R t}{2}\right), \quad \omega_R = \sqrt{\Delta^2 + |s_2|^2}$$

d) Solution

$$-\dot{c}_2(t) = \frac{-i s_2^*}{2} e^{-i \Delta t} \underbrace{c_1(t)}_{=1}$$

$$\begin{aligned} \dot{c}_2^{(H)}(t) &= \frac{-i s_2^*}{2} e^{-i \Delta t} \cdot 1 \\ &= \frac{-i s_2^*}{2} e^{-i \Delta t} \end{aligned}$$

Integrate from 0 to t, using $c_2^{(H)}(0) = 0$

$$\int \dot{c}_2^{(H)}(t) dt = \int_0^t \frac{-i s_2^*}{2} e^{-i \Delta t} dt$$

$$\begin{aligned} c_2^{(H)}(t) &= \int_0^t \frac{-i s_2^*}{2} e^{-i \Delta t} dt \\ &= \frac{-i s_2^*}{2} \int_0^t e^{-i \Delta t} dt \end{aligned}$$

8)

$$= \frac{\cancel{i\Omega} \times}{2} \frac{e^{-i\Delta t}}{\cancel{i\Delta t}} \Big|_0^t$$

$$= \frac{\Omega^*}{2\Delta} e^{-i\Delta t} \Big|_0^t$$

$$= \frac{\Omega^*}{2\Delta} (e^{-i\Delta t} - e^{-i\Delta \cdot 0})$$

$$= \frac{\Omega^*}{2\Delta} (e^{-i\Delta t} - e^0) = \frac{\Omega^*}{2\Delta} (e^{-i\Delta t} - 1)$$

The transition probability

$$P_{1 \rightarrow 2}^{(1)}(t) = |C_2^{(1)}(t)|^2$$

$$= \left| \frac{\Omega^*}{2\Delta} (e^{-i\Delta t} - 1) \right|^2$$

$$= \left| \frac{\Omega^*}{2\Delta} \right|^2 \cdot (e^{-i\Delta t} - 1)(e^{-i\Delta t} - 1)^*$$

$$= \left| \frac{\Omega^*}{2\Delta} \right|^2 (e^{-i\Delta t} - 1)(e^{i\Delta t} - 1)$$

$$= \left| \frac{\Omega^*}{2\Delta} \right|^2 (e^0 - e^{-i\Delta t} - e^{i\Delta t} + 1)$$

$$= \left| \frac{\Omega^*}{2\Delta} \right|^2 (2 + 1 - e^{-i\Delta t} - e^{i\Delta t})$$

$$= \left| \frac{\Omega^*}{2\Delta} \right|^2 (2 - 2\cos(\Delta t)) = \frac{|\Omega|^2}{4\Delta^2} 4\sin^2\left(\frac{\Delta t}{2}\right)$$

$$= \frac{|S|^2}{4\Delta^2} \sin^2\left(\frac{\Delta t}{2}\right)$$

$$= \frac{|S|^2}{\Delta^2} \sin^2\left(\frac{\Delta t}{2}\right)$$

Compare with the exact result

$$P_{2 \rightarrow 2}^{\text{exact}}(t) = \frac{|S|^2}{W_R^2} \sin^2\left(\frac{W_R t}{2}\right)$$

$$W_R = \sqrt{\Delta^2 + |S|^2}$$

$$= \sqrt{\Delta^2 + \frac{|S|^2}{\Delta^2}}$$

$$\approx |\Delta| \left(\Delta + \frac{|S|^2}{2\Delta^2} \right)$$

$$\approx |\Delta|$$

Then, $\frac{|S|^2}{W_R^2} \approx \frac{|S|^2}{\Delta^2}$

$$\sin^2\left(\frac{W_R t}{2}\right) \approx \sin^2\left(\frac{\Delta t}{2}\right)$$

$$P_{2 \rightarrow 2}^{\text{exact}}(t) \approx \frac{|S|^2}{\Delta^2} \sin^2\left(\frac{\Delta t}{2}\right)$$

$$= P_{2 \rightarrow 2}^{(h)}(t)$$

$$P_{2 \rightarrow 2}^{\text{exact}}(t) = P_{2 \rightarrow 2}^{(h)}(t)$$

$$P_{2 \rightarrow 2}^{\text{exact}}(t) \approx P_{2 \rightarrow 2}^{(h)}(t)$$

2) Given
 $H_0 = \sum_i \hbar \omega_i |i\rangle\langle i|$

$$V(t) = V_0 \cos(\omega t)$$

$$\langle \psi(0) | = |0\rangle$$

Required

$$\langle 3 | \psi_I^{(2)}(t) \rangle = 0 \quad \text{show this}$$

Solution

$$\begin{aligned} |\psi_I^{(1)}(t)\rangle &= |\psi(0)\rangle - \frac{i}{\hbar} \int_0^t dt V_I(t) |\psi(0)\rangle \\ &= |0\rangle - \frac{i}{\hbar} \int_0^t dt V_I(t) |0\rangle \end{aligned}$$

$$\begin{aligned} V_I(t) &= U_0^\dagger(t) V(t) U_0(t) \\ &= e^{i \frac{H_0 t}{\hbar}} V_0 \cos(\omega t) e^{-i \frac{H_0 t}{\hbar}} \end{aligned}$$

Then, $|\psi_I^{(2)}(t)\rangle = |0\rangle - \frac{i}{\hbar} \int_0^t dt e^{i \frac{H_0 t}{\hbar}} V_0 \cos(\omega t) e^{-i \frac{H_0 t}{\hbar}} |0\rangle$

Substituted all values

$$\begin{aligned} \langle 3 | \psi_I^{(2)}(t) \rangle &= \underbrace{\langle 3 | 0 \rangle}_{=0} - \frac{i}{\hbar} \int_0^t dt \langle 3 | e^{i \frac{H_0 t}{\hbar}} V_0 \cos(\omega t) e^{-i \frac{H_0 t}{\hbar}} | 0 \rangle \\ &= 0 - \frac{i}{\hbar} \int_0^t dt e^{i \frac{E_3 t}{\hbar}} \langle 3 | V_0 \cos(\omega t) | 0 \rangle e^{-i \frac{E_0 t}{\hbar}} \\ &= -\frac{i}{\hbar} \int_0^t dt e^{i \frac{(E_3 - E_0) t}{\hbar}} \langle 3 | V_0 \cos(\omega t) | 0 \rangle \\ &= -\frac{i}{\hbar} \int_0^t dt e^{i \frac{(E_3 - E_0) t}{\hbar}} \cos(\omega t) \underbrace{\langle 3 | V_0 | 0 \rangle}_{=0} \\ &= -\frac{i}{\hbar} \int_0^t dt e^{i \frac{(E_3 - E_0) t}{\hbar}} \cos(\omega t) \cdot 0 \end{aligned}$$

22)

$$= - (0)$$

$$= 0$$

$$\langle 3 | \psi_1^{(1)}(t) \rangle = 0$$

b) Solution

$$\begin{aligned} |\psi_1^{(2)}(t)\rangle &= |\psi(0)\rangle - \frac{i}{\hbar} \int_0^t dt_1 V_I(t_1) |\psi(0)\rangle + \left(\frac{-i}{\hbar}\right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 V_I(t_1) V_I(t_2) |\psi(0)\rangle \\ &= |\psi(0)\rangle - \frac{i}{\hbar} \int_0^t dt_1 V_I(t_1) |\psi(0)\rangle + \left(\frac{-i}{\hbar}\right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 V_I(t_1) V_I(t_2) |\psi(0)\rangle \end{aligned}$$

$$\begin{aligned} \langle 3 | \psi_2^{(2)}(t) \rangle &= \langle 3 | \left(|\psi(0)\rangle - \frac{i}{\hbar} \int_0^t dt_1 V_I(t_1) |\psi(0)\rangle + \left(\frac{-i}{\hbar}\right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 V_I(t_1) V_I(t_2) |\psi(0)\rangle \right) \\ &= 0 + \langle 3 | \left(\left(\frac{-i}{\hbar}\right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 V_I(t_1) V_I(t_2) |\psi(0)\rangle \right) \end{aligned}$$

$$= -\frac{1}{\hbar^2} \int_0^t dt_1 \int_0^{t_1} dt_2 \langle 3 | (V_I(t_1) V_I(t_2)) | 2 \rangle$$

$$= -\frac{1}{\hbar^2} \int_0^t dt_1 \int_0^{t_1} dt_2 \langle 3 | V_I(t_1) V_I(t_2) | 2 \rangle$$

$$= -\frac{1}{\hbar^2} \int_0^t dt_1 \int_0^{t_1} dt_2 \langle 3 | V_2(t_1) I V_2(t_2) | 2 \rangle$$

12)

$$\begin{aligned}
 \langle 3 | V_2(t_1) \cdot V_2(t_2) | 1 \rangle &= \underbrace{\langle 3 | V_2(t_1) (12741 + 127421 + 132281) | 1 \rangle}_{=0} \underbrace{V_2(t_2) | 1 \rangle}_{=0} \\
 &= \\
 &= \langle 3 | V_2(t_1) (127421) V_2(t_2) | 1 \rangle \\
 &= \langle 3 | V_2(t_1) | 2 \rangle \langle 2 | V_2(t_2) | 1 \rangle
 \end{aligned}$$

generally

$$V_I(t) = e^{iH_0 t / \hbar} V(t) e^{-iH_0 t / \hbar}, \quad \hbar = 1$$

$$V_I(t) = e^{iH_0 t} V(t) e^{-iH_0 t}$$

$$= e$$

$$\langle i | U_I(t) | j \rangle = \langle i | e^{iH_0 t / \hbar} V(t) e^{-iH_0 t / \hbar} | j \rangle$$

$$= e^{iE_i t / \hbar} \langle i | V(t) \cos(\omega t) | j \rangle e^{-iE_j t / \hbar}$$

$$= e^{iE_i t / \hbar} \underbrace{\langle i | V(t) | j \rangle}_{=V_{ij}} e^{-iE_j t / \hbar} \cos(\omega t)$$

$$= V_{ij} \cos(\omega t) e^{i \underbrace{(E_i - E_j)}_{= \hbar \omega} t / \hbar}$$

$$= V_{ij} \cos(\omega t) e^{i \omega t}$$

$$= V_{ij} \left[\frac{e^{i \omega t} + e^{-i \omega t}}{2} \right] e^{i \omega t}$$

$$= \frac{V_{13}}{2} \left[e^{i\omega_1 t + i\omega_{13} t} + e^{-i\omega_1 t + i\omega_{13} t} \right]$$

$$= \frac{V_{13}}{2} \left[e^{i(\omega_{13} + \omega)t} + e^{i(\omega_{13} - \omega)t} \right]$$

where, $\Delta_{13} = \omega_{13} - \omega$

then

$$= \frac{V_{13}^0}{2} \left[e^{i(\omega_{13} + \omega)t} + e^{i(\Delta_{13}^0)t} \right]$$

Using the Assumption $\omega, \omega_{21}, \omega_{23} \gg |\Delta_{21}|, |\Delta_{23}|$

Keep only the slowly varying (near-resonant) term

$$\langle 2 | U_2(t) | 1 \rangle \approx \frac{V_{21}}{2} e^{i\Delta_{21}t}$$

$$\langle 3 | U_2(t) | 2 \rangle \approx \frac{V_{32}}{2} e^{-i\Delta_{23}t}$$

$$\begin{aligned} \langle 3 | U_2^{(2)}(t) \rangle &\approx - \frac{V_{32}}{2} \frac{V_{21}}{2} \int_0^t dt_1 \int_0^{t_1} dt_2 e^{-i\Delta_{23}t_1} e^{i\Delta_{21}t_2} \\ &= - \frac{V_{32} V_{21}}{4} \int_0^t dt_1 e^{-i\Delta_{23}t_1} \underbrace{\int_0^{t_1} dt_2 e^{i\Delta_{21}t_2}}_{\frac{e^{i\Delta_{21}t_1} - 1}{i\Delta_{21}}} \\ &= - \frac{V_{32} V_{21}}{4} \int_0^t dt_1 e^{-i\Delta_{23}t_1} \left(\frac{e^{i\Delta_{21}t_1} - 1}{i\Delta_{21}} \right) e^{i\Delta_{21}t_1} \end{aligned}$$

(4)

$$\begin{aligned}
&= -\frac{V_{32}V_{21}}{4i\Delta_{21}} \int_0^t dt_1 \left[e^{i(\Delta_{21}-\Delta_{23})t_1} - e^{-i\Delta_{23}t_1} \right] \\
&= -\frac{V_{32}V_{21}}{4i\Delta_{21}} \left[\int_0^t e^{i(\Delta_{21}-\Delta_{23})t_1} dt_1 - \int_0^t e^{-i\Delta_{23}t_1} dt_1 \right] \\
&= -\frac{V_{32}V_{21}}{4i\Delta_{21}} \left[\frac{e^{i(\Delta_{21}-\Delta_{23})t} - 1}{i(\Delta_{21}-\Delta_{23})} - \frac{1 - e^{-i\Delta_{23}t}}{i\Delta_{23}} \right] \\
&= \frac{V_{32}V_{21}}{4\Delta_{21}} \left[\frac{e^{i(\Delta_{21}-\Delta_{23})t} - 1}{\Delta_{21}-\Delta_{23}} - \frac{1 - e^{-i\Delta_{23}t}}{\Delta_{23}} \right]
\end{aligned}$$

Use, that,

$$\Delta_{21} - \Delta_{23} = (\omega_{21} - \omega) - (\omega_{23} - \omega)$$

$$= \omega_{21} - \omega_{23}$$

$$= \underline{\underline{\omega_{31}}}$$

$$\langle 3 | \psi_I^{(2)}(t) \rangle = \frac{V_{32}V_{21}}{4\Delta_{21}} \left[\frac{e^{i\omega_{31}t} - 1}{\omega_{31}} - \frac{1 - e^{-i\Delta_{23}t}}{\Delta_{23}} \right]$$

Apply the above limits and drop fast / higher-order term

$$|\Delta_{21}|, |\Delta_{23}| \gg \omega_{31}$$

$$\Delta_{21} \approx \Delta_{23} \approx \Delta$$

$$\langle 3 | \psi_I^{(2)}(t) \rangle \approx \frac{V_{32}V_{21}}{4\Delta} \frac{e^{i\omega_{31}t} - 1}{\omega_{31}} //$$

(5)

(1) Solution

$$\begin{aligned}
 P_{2 \rightarrow 3}(t) &= |\langle 3 | \psi(t) \rangle|^2 = \left| \frac{V_{32} V_{21}}{4\Delta} \right|^2 \left| \frac{e^{i\omega_{31}t} - 1}{\omega_{31}^2} \right|^2 \\
 &= \left| \frac{V_{32} V_{21}}{4\Delta} \right|^2 \frac{|e^{i\omega_{31}t} - 1|^2}{\omega_{31}^2} = \left| \frac{V_{32} V_{21}}{4\Delta} \right|^2 \frac{2(1 - \cos(\omega_{31}t))}{\omega_{31}^2} \\
 &= \left| \frac{V_{32} V_{21}}{4\Delta} \right|^2 \frac{4 \sin^2\left(\frac{\omega_{31}t}{2}\right)}{\omega_{31}^2} \\
 &= 4 \left| \frac{V_{32} V_{21}}{4\Delta} \right|^2 \frac{\sin^2\left(\frac{\omega_{31}t}{2}\right)}{\omega_{31}^2}
 \end{aligned}$$



Consider a two level system with a direct constant coupling

$$V_{\text{eff}} = V (|3\rangle\langle 2| + |2\rangle\langle 3|)$$

$$\langle 3 | V_{\text{eff}}(t) | 2 \rangle = V e^{i\omega_{31}t}$$

$$\begin{aligned}
 \langle 3 | \psi(t) \rangle &= -i \int_0^t dt \langle 3 | V_{\text{eff}}(t) | 2 \rangle \\
 &= V \frac{e^{i\omega_{31}t} - 1}{i\omega_{31}}
 \end{aligned}$$

$$P_{2 \rightarrow 3}^{\text{(direct)}}(t) = |V|^2 \frac{|e^{i\omega_{31}t} - 1|^2}{\omega_{31}^2} = 4|V|^2 \frac{\sin^2\left(\frac{\omega_{31}t}{2}\right)}{\omega_{31}^2}$$

Therefore, $V_{\text{eff}} = \frac{V_{32} V_{21}}{4\Delta}$

Effective coupling

$$V_{eff} = \frac{V_m V_{02}}{\Delta \omega}$$

b) Hydrogen atom in a capacitor

a) given

$$\vec{E}(t) = E_0 e^{-\beta t} \theta(t) \hat{e}_z$$

Solution

$$\vec{E}(t) = E_0 e^{-\beta t} \theta(t) \hat{e}_z$$

$$\begin{aligned} V(t) &= -e \vec{E}(t) \cdot \vec{z} \\ &= -e (E_0 e^{-\beta t} \theta(t) \hat{e}_z) \cdot \vec{z} \\ &= -e E_0 e^{-\beta t} \theta(t) z. \end{aligned}$$

First order time-dependent perturbation theory

$$\begin{aligned} C_f^{(1)}(t) &= -\frac{i}{\hbar} \int_0^t dt' e^{i\omega_f t'} \langle f | V(t') | i \rangle, & \text{where} \\ &= -\frac{i}{\hbar} \int_0^t dt' e^{i\omega_f t'} \langle 2s | V(t') | 2s \rangle & \omega_f = \frac{E_f - E_i}{\hbar} \\ &= -\frac{i}{\hbar} \int_0^t dt' e^{i\omega_f t'} \langle 2s | (-e E_0 e^{-\beta t'} \theta(t') z) | 2s \rangle & |f\rangle = |2s\rangle \\ &= -\frac{i}{\hbar} \int_0^t dt' e^{i\omega_f t'} (-e E_0 e^{-\beta t'} \theta(t')) \underbrace{\langle 2s | z | 2s \rangle}_{=0} & |i\rangle = |2s\rangle \end{aligned}$$

$$c_{25}^{(1)}(t) = \frac{-i}{2} \int_0^t dt e^{i\omega_{25} t} e^{i\omega_{25} t} (-e E_0 e^{-\gamma t}) \cdot 0$$

$$= \underline{\underline{0}}$$

The transition probability

$$P_{25 \rightarrow 25}(t) = |c_{25}^{(1)}(t)|^2$$

$$= |0|^2$$

$$= \underline{\underline{0}}$$

b) The 2p-state

Solution

$$\langle 2p_0 | V(t) | 2s \rangle = -e E_0 e^{-\gamma t} \langle 2p_0 | 2s \rangle$$

so,

$$c_{2p}^{(1)} = \frac{-ie E_0}{\hbar} \langle 2p_0 | 2s \rangle \int_0^t dt e^{(i\omega_{2p,2s} - \gamma)t}$$

$$= \frac{1}{\beta - i\omega}$$

$$= \frac{-ie E_0}{\hbar} \langle 2p_0 | 2s \rangle \cdot \frac{1}{\beta - i\omega} = \underline{\underline{\frac{-ie E_0 \langle 2p_0 | 2s \rangle}{\hbar \cdot \beta - i\omega_{2p,2s}}}}$$

$$\frac{c_{2p0}^{(1)}}{\beta - i\omega} = -\frac{e E_0}{\hbar \omega_{2p,2s}} \langle 2p_0 | 2s \rangle$$

$$P_{2s \rightarrow 2p}(t) = |c_{2p0}^{(1)}|^2$$

$$= \frac{e^2 E_0^2}{\hbar^2 \omega_{2p,2s}^2} |\langle 2p_0 | 2s \rangle|^2$$

fine dipole matrix element $\langle 2p0 | z | 1s \rangle$

$$\psi_{1s}(r, \theta, \phi) = \frac{1}{\sqrt{\pi} a_0^3} e^{-r/a_0}$$

$$\psi_{2p0}(r, \theta, \phi) = \frac{1}{4\sqrt{2\pi} a_0^3} \left(\frac{r}{a_0}\right) e^{-r/2a_0} \cos\theta$$

$$\begin{aligned} \langle 2p0 | z | 1s \rangle &= \int d^3r \psi_{2p0}^*(r, \theta, \phi) (r \cos\theta) \psi_{1s}(r, \theta, \phi) \\ &= \int r^2 \sin\theta dr d\theta d\phi \psi_{2p0}^*(r, \theta, \phi) (r \cos\theta) \psi_{1s}(r, \theta, \phi) \end{aligned}$$

where

$$\frac{4\pi}{3} \equiv \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta = \text{Angular part}$$

radial part

$$\frac{4!}{a^5} = \int_0^\infty r^4 e^{-ar} dr, \text{ where, } a = \frac{3}{2a_0}$$

I obtain,

$$\int_0^\infty dr r^4 e^{-3r/(2a_0)} = \frac{4!}{\left(\frac{3}{2a_0}\right)^5}$$

$$\langle 2p0 | z | 1s \rangle = \frac{128\sqrt{2}}{243\sqrt{\pi}} a_0$$

$$|\langle 2p0 | z | 1s \rangle|^2 = \frac{32768}{59049\pi} a_0^2$$

$$P_{2p \rightarrow 1s}(t) \approx P(t=0) = \frac{e^2 E_0^2}{4\hbar^2 \omega_{2p,1s}^2} \frac{32768}{59049\pi} a_0^2$$